Module 3: Bison reintroduction and plant biomass

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Introduction

For many years, the K'Avi tribe has had cattle grazing pastures throughout their land. Cattle have been a useful agricultural resource, but they are not native to the K'Avi's region. In the past year, tribal agriculture managers have noticed a decreased vitality in many of their grazing lands causing less regrowth and a decrease in native plant species. This is most likely due to the cattle, as they are non-specific grazers and are a non-native species. Tribal agriculture managers are considering reintroducing bison to the lands for both ecological and cultural reasons. Bison are native to this region and have been observed to increase native grass species richness by 103% in comparison to cattle! Reintroducing bison will also re-establish the K'Avi's cultural and historical connections to the land, and can reaffirm food sovereignty and provide additional economic benefits. The managers are hoping to repurpose the cattle grazing lands to become bison grazing lands to foster a more sustainable agricultural practice for the region and to ensure the preservation of the native grasslands.

In collaborations with numerous tribal governments, K'Avi agriculture managers are hoping to transfer a herd of 200 bison for reintroduction, which would allow for genetic diversity within the herd and opportunities for sustainable hunting.

To reintroduce bison, the land must have enough available sustenance for the bison to graze and maintain a healthy lifestyle. The tribal agriculture managers are considering three sites for reintroduction and have produced a dataset (biomass.csv) with the annual plant biomass (i.e., the amount of food available) at each site. They also used GIS to calculate the area of each site in hectares (one hectare is roughly equivalent to two football fields), producing the dataset "bison_sites.csv". These data will help determine which sites are viable for bison reintroduction! While the hope is to eventually reintroduce bison herds to all of these sites, K'Avi agricultural managers must currently choose a single site to place this first herd of 200 bison.

Learning Objectives for Module 3:

1. Learn to combine separate data sources into a single, cohesive dataset
2. Group data by specific categories (in this case field sites) to summarize and explore side-by-side comparisons
3. Standardize and convert different units of measurement to make data easier to interpret and compare
4. Evaluate multiple environmental variables simultaneously for managing decisions on reintroducing native species to their habitats

DAY 1

Lesson 3.1: Understanding Site Biomass Data

In this lesson we will explore plant biomass data and summarize it by site. Quadrats, which are $0.25m^2$ squares used to randomly collect data, were used because the sites are too large to actually measure the entire area. We will explore the biomass data by quadrat and by site, using a boxplot and summary functions.

Getting set up

First we will load packages and data. Similar to previous modules, the only package we will require here is `tidyverse`.

About the datasets:

`biomass` data frame includes the following columns: - `Site` : ID of site that is under consideration for bison reintroduction - `Quadrant` : a 0.25 square meter area over which data is sampled - `Biomass` : amount of annual plant biomass, in grams per $0.25m^2$

`sites` data frame includes the following columns: - `Site` : ID of site that is under consideration for bison reintroduction - `Hectares` : estimation of the size of each site, in hectares

Load in tidyverse

```
library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✔ dplyr     1.2.1     ✔ readr     2.2.0
## ✔ forcats   1.0.1     ✔ stringr   1.6.0
## ✔ ggplot2   4.0.3     ✔ tibble    3.3.1
## ✔ lubridate 1.9.5     ✔ tidyr     1.3.2
## ✔ purrr     1.2.2
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts
to become errors
```

Load in Biomass Data and Site Datasets

```
biomass <- read.csv("biomass.csv", header = TRUE, sep = ',')

sites <- read.csv("bison_sites.csv", header = TRUE, sep = ',')
```

Previously, we learned to work with data across multiple sites by having each site as a separate column. This approach allowed us to easily graph data from multiple sites over time. Here, since we will not be looking at data over time, we will be comparing data between multiple sites in a different way, with all of the sites in a single column. View the “biomass” dataset to see this structure.

💻 Use one of the following functions to preview both data sets: `View()`, `head()`, or `str()`

```
# preview your data frames using View() or head() or str()
```

In the bimass data frame, we have annual plant biomass data from 50 $0.25m^2$ quadrats (small squares within which data is sampled), for each site. That means each site has a distribution of values representing plant biomass per $0.25m^2$. We cam compare these distributions with the boxplot below, however, because the data frame has two columns that could be categorical (`Site` and `Quadrat`), we need to specify how to group it.

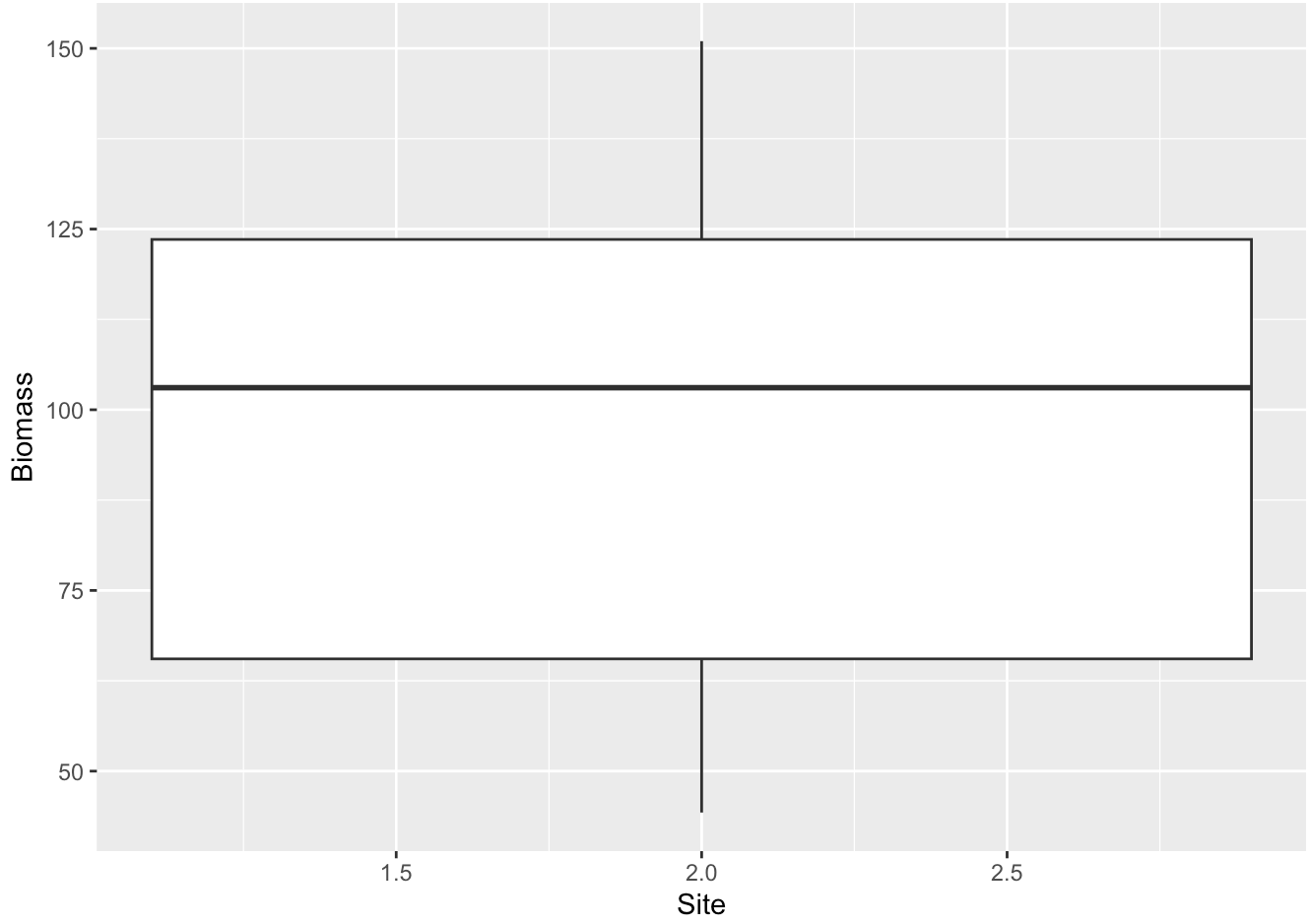
💻 Run the code given in the next chunk and see the Warning message and output.

Then, edit the line so that it includes `group=Site` inside the `aes()` . Rerun and look at the output.

```
ggplot(biomass, aes(x=Site, y=Biomass)) + geom_boxplot()
```

```
## Warning: Orientation is not uniquely specified when both the x and y aesthetics are
## continuous. Picking default orientation 'x'.
```

```
## Warning: Continuous x aesthetic
## ℹ did you forget `aes(group = ...)`?
```



Reminder to edit the previous code so that it includes `group=Site` inside the `aes()` . Rerun and look at the output.

Grouping data

With the data for all of the sites in the same columns, if we took the summary statistics for a column, it would take the summary statistics for all of the sites together, rather than the summary statistics for each site. In order to run code for each site, we can use the `group_by()` function, which is part of the tidyverse. The `group_by()` function tells R to group rows of data by their shared values in a given column. A sample of this code can be seen below:

```
biomass_grouped <- dplyr::group_by(biomass, Site)
#A breakdown of the components of this code: group_by(name of the dataset, column with v
alues to group by)
#In this code, we are saying to group data in the "biomass" dataset by the values (eithe
r 1, 2, or 3) in the "Site" column. This will group all of the data in the Site, Quadra
t, and Biomass columns according to the value in the "Site" column.
head(biomass_grouped)
```

```
## # A tibble: 6 × 3
## # Groups:   Site [1]
##    Site Quadrat Biomass
##   <int>   <int>   <dbl>
## 1     1       1    55.5
## 2     1       2    58.2
## 3     1       3    72.5
## 4     1       4    60.6
## 5     1       5    61.0
## 6     1       6    73.7
```

The next sample displays what the summary of the data looks like, with and without grouping:

```
# group by Site
biomass_grouped <- dplyr::group_by(biomass, Site)

# compare the output
paste("without groups:")
```

```
## [1] "without groups:"
```

```
summarize(biomass, mean(Biomass))
```

```
##   mean(Biomass)
## 1      96.88813
```

```
# once grouped, you can apply summary functions that will maintain groupings
paste("with groups:")
```

```
## [1] "with groups:"
```

```
summarize(biomass_grouped, mean(Biomass))
```

```
## # A tibble: 3 × 2
##    Site `mean(Biomass)`
##   <int>           <dbl>
## 1     1            60.3
## 2     2           103.
## 3     3           127.
```

```
# this gives us the average plant biomass within a quadrant for each site
```

A Note about tibbles

We can see that this code chunk also just visualized the first section of our data as a table. These tables we produce are called “tibbles” in tidyverse. They are another, simpler way to view the data, and are basically the same as a data frame. If you want more information, read more here. (https://tibble.tidyverse.org/)

A Note About dplyr

The dplyr package, which is part of the tidyverse, provides numerous unique tools to easily create, edit, and manipulate data, which is often necessary when your raw data is not perfectly set up to answer the questions you are asking. Mastering these tools will give you data management skills relevant and desirable in both academic and professional settings. In this module, we will only discuss a few of the potential data management functions available, but you are encouraged to explore all of the tools and make decisions for the appropriate ones to use as you see fit! A “cheat sheet” explaining all of these functions can be found here: dplyr cheat sheet (https://www.rstudio.com/wp-content/uploads/2015/02/data-wrangling-cheatsheet.pdf)

Now, let’s look again at the data that we have: we want to figure out which sites have enough food (plant biomass) to support a herd of 200 reintroduced bison. To figure this out, the first step is to calculate the total amount of plant biomass at each site. However, as the “sites” dataset shows us, these sites are far too large to manually count up the plant biomass for the entire site. Fortunately, this is not necessary. Instead, tribal researchers collected annual plant biomass from fifty 0.25 square meter quadrats (small squares within which data is sampled) at each site, which can be used to estimate the total biomass at each site.

While this collection method is actually feasible, this dataset doesn’t quite yet give the information the tribal agriculture managers are looking for, which is the total biomass at each site.

Combine group_by() and summarize()

What we have right now is the annual biomass in fifty 0.25 square meter quadrats at each site. How can we turn these fifty values into an estimate of the total biomass at each site?

We repeat the steps above, but combine them into one step using the pipe function (`%>%`). (note: Even though this code chunk produced a 3 x 2 tibble showing us average biomass at each site, this did not change the “biomass” dataset, and we can see that this tibble is not in our Global Environment, and does not actually have a name. This is because we did not give this tibble a name and assign it to a new dataset (using the `<-`))

```
biomass %>%
  group_by(Site) %>%
  summarize(avg_biomass=mean(Biomass))
```

```
## # A tibble: 3 × 2
##    Site avg_biomass
##   <int>       <dbl>
## 1     1        60.3
## 2     2       103.
## 3     3       127.
```

Side note: Even though this code chunk produced a 3 x 2 tibble showing us average biomass at each site, this did not change the "biomass" dataset, and we can see that this tibble is not in our Global Environment, and does not actually have a name. This is because we did not give this tibble a name and assign it to a new dataset (using the <-).

🧠✍️ Class Questions:

- What site has the highest average plant biomass in its quadrats? What might this tell us about the viability of these sites for bison reintroduction?
- What's the problem with having data in two different units? What should we change the units to?

Lesson 3.2: Unit Conversions

This focus of this lesson is unit conversions, and using `mutate()` to create new columns in our data frame.

As a reminder: We take the average quadrat biomass at each site because due to the large size of the sites, it would be extremely time consuming and expensive to get an actual total biomass of each site. This average gives us an idea of the average biomass in each of the 50 quadrats where data was taken. This data is currently in $g/0.25m^2$, but the site area data has been given in different units– hectares. Bison also consume a lot of plants, with a single bison needing about 21,900 kg (1 kg = 1000g) of biomass a year to live a healthy life. So, we have data in different units from one another. Fortunately, we can easily do math in R to create new variables with updated units.

Here are the unit conversions to from g/0.25m² to kg/ha: 1. 1 hectare = 10,000 m² 2. 1 kg = 1000 g 3. 1 kg/ha = 10g/m² 4. 1 g/0.25m² = 4g/m² 5. 4g/m² multiplied by 10 = 40kg/ha 6. Conversion of g/0.25m² to kg/ha is to multiply by 40

To do this conversion and add it to our data, we need to make a new column with the converted units. To make a new column in a dataset, we can use the `mutate()` function in tidyverse. `mutate()` creates new columns that are functions of existing variables. It can also modify and delete columns, but we won't do either of those today. To make sure that this does not alter the original data, we will make a copy of the biomass dataset with the new column and assign it to a new dataset (using <-) called "biomass_kgha". Since this is just adding a new column to the original data, you can also apply these changes to "biomass" and not make a new dataset by doing "biomass <-" instead. Just make sure to be careful when making changes to any dataset!

Converting units to kg/ha

```
biomass_kgha <- mutate(biomass, kg_ha = Biomass * 40)
#the various parts of the mutate() function: mutate(name of dataset, new column = functi
on of existing column)

head(biomass_kgha) #preview the dataframe with the new column
```

```
##   Site Quadrat  Biomass    kg_ha
## 1    1       1 55.51619 2220.648
## 2    1       2 58.15858 2326.343
## 3    1       3 72.46967 2898.787
## 4    1       4 60.56407 2422.563
## 5    1       5 61.03430 2441.372
## 6    1       6 73.72052 2948.821
```

This new dataset `biomass_kgha` now has a new column that converts the biomass in g/m^2 into kg/ha. The column "kg_ha" will be the new column we will call when we are looking at the biomass data because its now in the correct units!

We should take the average biomass per quadrat again, but this time with the converted units.

Find the average biomass in kg/ha

```
biomass_avg <- biomass_kgha %>%
  group_by(Site) %>%
  summarize(avg_biomass_kg_ha=mean(kg_ha))
biomass_avg
```

```
## # A tibble: 3 × 2
##    Site avg_biomass_kg_ha
##   <int>             <dbl>
## 1     1             2411.
## 2     2             4117.
## 3     3             5098.
```

We have just used two different ways of manipulating and creating data in R: the `summarize()` and `mutate()` functions.

🧠✍️ Class Questions:

- What is the difference between using `summarize()` and `mutate()` ?
- How could you combine the steps above for unit conversion and summarizing using a pipe (`%>%`)?

If you'd like, play around with both `mutate()` and `group_by() %>% summarize()` later on in this module to get a clearer idea of how they differ from one another.

Join datasets of average plant biomass and site area

There is another dataset the tribal agriculture managers have with the sizes of each site in hectares, called “sites”, which we loaded in with biomass data at the beginning of class. We are going to merge this “sites” dataset with our `biomass_avg` dataset, which has the average biomass at each site (kg per hectare), into one dataset using the function left_join(), which we learned in the Water Module. We can do this because both datasets have matching columns called Site we can use as the key.

```
site_biomass <- left_join(sites, biomass_avg, by = "Site") #left_join(dataset 1, dataset 2, by = "shared column"); Join matching rows from dataset 2 to dataset 1 using their shared column "Site".
site_biomass
```

```
##   Site Hectares avg_biomass_kg_ha
## 1    1     1700          2411.009
## 2    2     1300          4117.127
## 3    3      900          5098.440
```

Class Questions

- How does merging the datasets help us see the data?
- What can this new tibble tell us about the quantity of biomass in each site?
- Just from viewing this dataset, do you have a hypothesis of which site might be most suitable to reintroduce bison?

To get a rough estimate of the total available biomass in each site we should multiply the average biomass per quadrat (kg/ha) by the hectares of each site (ha). This will give us a total kg of biomass at each site. This should be added as a new column to the site_biomass tibble.

$$total_{a}vailable_{b}iomass(kg) = area(ha) * biomass(kg/ha)$$

Estimate total available biomass

```
total_available <- mutate(site_biomass, total_biomass = Hectares * avg_biomass_kg_ha)
total_available
```

```
##   Site Hectares avg_biomass_kg_ha total_biomass
## 1    1     1700          2411.009       4098716
## 2    2     1300          4117.127       5352265
## 3    3      900          5098.440       4588596
```

Class Questions

- What can these totals tell us about each site? Which site do you think is best suited for bison reintroduction? Why?

Estimate Bison capacity

Bison eat a lot of plants! One bison needs about 21,900 kg of biomass a year to live a healthy life. Since we have the total kg of biomass for each site now, we have an idea of which site might be best to introduce the bison. Bison are social animals and live in herds. It is important to the tribal agriculture managers to ensure that they can sustain a full herd population of at least 200 bison.

Using `mutate()`, how many bison can each of these sites maintain?

```
# fill in the mutate function to create a new column of number of bison
max_bison<-mutate(total_available, max_bison=total_biomass/21900)
max_bison
```

```
##   Site Hectares avg_biomass_kg_ha total_biomass max_bison
## 1    1     1700          2411.009       4098716  187.1560
## 2    2     1300          4117.127       5352265  244.3956
## 3    3      900          5098.440       4588596  209.5249
```

Class Questions

- Which site(s) would be able to maintain a healthy bison population (around 200 Bison)?
- If you were to share these results with the local community and to other agriculture managers, how might you visualize this data to show which site might be best?

Data visualization of max bison per site

💻 Your turn to create a plot! Here is link to ggplot cheatsheet (https://rstudio.github.io/cheatsheets/html/data-visualization.html) for exploring different plots.

```
#Write ggplot code here to make a graph that visualizes the maximum viable bison populat
ion at each site!
```

DAY 2

Lesson 3.3: Nitrogen data for grazing bison

From the first two modules, we have identified multiple sites that have enough biomass to sustain a healthy herd of 200 bison for reintroduction! This is great, but upon further investigation, tribal agriculture managers realize that the total biomass at each site is not the sole determinant of how many bison each site can support.

Today, you will be tasked with using the code from Lesson 3.2 to investigate other properties of each site: the amount of available nitrogen for bison to consume and/or the amount of biomass available for bison after accounting for each sites' existing elk populations, and how they each may effect the total number of bison that each site can support.

You are encouraged to first work with whichever dataset (Nitrogen or Elk) you find more interesting, and if you have enough time to then work on the other dataset. The code we used from last class, where we learned how to manipulate and create data, can be directly applied to address these additional factors.

Nitrogen data

While the physical amount of biomass might be enough at multiple sites to support a herd of 200 bison, the quality of the available forage is also important to consider. Since bison are generalist foragers, they'll eat almost anything, but they need to make sure that they consume 73,000 grams of nitrogen (or "N") a year. Luckily, the tribal agriculture managers have another dataset that has the nitrogen content per quadrat at each site, following the same data structure as the "biomass" dataset from last class. All nitrogen values are in g/0.25 m2, while the site data is in hectares, so you will want to convert your units so that they match.

If you want your units of N to be in kg/ha, the conversion is to multiply your nitrogen data by 40.

Import and summarize N data

```
nitrogen <- read.csv("Nitrogen.csv", header = TRUE, sep = ',')
```

Since the data is given in quadrats again, how would we find the total amount of nitrogen available at each site for the bison? Look at the code from last class for reference!

Estimate total nitrogen per site

💻 We will go through the same steps as the plant biomass data:

1. create a new column in the `nitrogen` dataframe that converts the units in `Nitrogen` column to kg/ha, and call this column `N_kg_ha` .
2. Then summarize the average N_kg_ha per site.
3. Join with the `sites` data frame
4. Calculate the total N per site estimation
5. Estimate the number of bison the site could sustain, considering a single bison consumes 73,000 grams of nitrogen/year

```
# 1. create a new column in the `nitrogen` dataframe that converts the units in `Nitroge
n` column to kg/ha, and call this column `N_kg_ha`.
# hint: use the mutate function to convert units to kg/ha
```

```
# 2. Then summarize the average N_kg_ha per site.
# hint: use group_by(site) and summarize together
```

```
# 3. Join with the `sites` data frame
# hint use the join() function to join nitrogen data frame with sites
```

```
# 4. Calculate the total N per site estimation
# hint: use mutate to create a column that is total N per site
```

```
# 5. Estimate the number of bison the site could sustain, considering a single bison con
sumes 73,000 grams of nitrogen/year
# hint: use mutate again to convert from nitrogen per site to bison per site
```

Class Question

- Does this change your decision for which site would be best for reintroduction? Why or why not?
- What additional step could you take to consider both Nitrogen and plant biomass at the same time?

Visualizing N data

Now that we have the amount of bison that each site's nitrogen can support, tribal agriculture managers want a visual representation of this data to share with the K'Avi community. This can be a tibble, graph, etc, but they want something that clearly communicates the data with understandable units and a title. Make it pretty! The most effective communication is clear and concise!

💻 Use ggplot to create a visualization that includes Nitrogen per site. Here is link to ggplot cheatsheet (https://rstudio.github.io/cheatsheets/html/data-visualization.html) for exploring different plots.

```
# use ggplot here
```

Class Questions

- Why do you believe this is the best way to present the data?
- Does this change which sites we should introduce the bison to?
- Why is it important to consider outside factors when working with data?
- Are there any other factors you would consider when looking to reintroduce a species into its native habitat?

Lesson 3.4: Plant biomass for elk populations

Even though there might be multiple sites with enough biomass to support a large herd of bison, it is not plausible that the bison will eat 100% of the forage available. There are elk populations present in the prairies that must be accounted for to ensure that both groups have enough biomass to sustain their populations.The present elk population must be able to coexist with the reintroduced bison population. Elk require 2,158 kg/year of plant biomass. As a reminder, bison require 21,900 kg/year of plant biomass.

Import elk dataset

```
elk <- read.csv("elk.csv", header = TRUE, sep = ',')

# add line here to preview elk data
```

Class Questions

- How many elk are there at each site?
- Considering the elk populations and the total biomass (from last class) at each site, do you expect the suitable sites for bison reintroduction to change? Why or why not?

If each elk requires 2,158 kg/year of plant biomass, how much biomass are the elk consuming at each site? What code would you use to figure this out?

```
# hint: if we know how many elk there are, we can convert to how much plant biomass is c
onsumed

# name the new column 'plant_biomass'
```

Class Questions

- Are these herds of elk consuming more or less biomass than the herd of 200 bison would?
- What does this tell us about the effect they'll have on the site we reintroduce bison to?
- How would we go about determining the total biomass available at each site when the elk consumption is taken into consideration?

Find new total biomass with Elk

```
# first join the dataframe of max_bison, which includes a column for total biomass, with
the elk dataframe
bison_and_elk = inner_join(max_bison, elk, by='Site') %>%
  # then create a new column with the new total biomass available with elk consumption
  mutate(total_biomass_w_elk = total_biomass - plant_biomass)

# preview the new dataframe
head(bison_and_elk)
```

💻 Next, use ggplot and `geom_col()` to compare the new total biomass with elk across sites (`total_biomass_w_elk`).

```
# insert ggplot here
```

Class Questions

- Which site has the most biomass available to bison, after accounting for biomass consumed by elk? Does considering biomass consumption by elk change the site with the most available biomass?

Max sustainable bison amount after elk consumption

It's great to have an understanding of just how much biomass is available for the bison with the elk in consideration. As tribal agriculture managers begin the process of reintroduction, they once again would like to know the maximum number of bison they can introduce to each site, after considering the biomass already being consumed by elk populations. Ideally, they would be able to introduce a herd of at least 200 bison to ensure a healthy and stable population.

Divide post-elk biomass by bison biomass requirements to get new max viable bison population. Recall that one bison needs about 21,900 kg of biomass.

```
# insert code here - hint, use mutate
```

Communicate this data in a fashion that would communicate well to the K'Avi community. This can be a table, graph, etc, but they want something that clearly communicates which sites have the biomass to support the reintroduction of 200 bison, with understandable units and a title. Make it pretty! The most effective

communication is clear and concise!

💻 Try using ggplot to create a visualization, different from the previous column chart.

```
# insert ggplot here
```

Class Questions

- Why do you believe this is the best way to present the data?
- Does this change which sites we should introduce the bison to?
- Why is it important to consider outside factors when working with data?
- Are there any other factors you would consider when looking to reintroduce a species into its native habitat?

Optional: If you have worked through both the nitrogen and elk datasets, and are looking for more to do, you can see whether there is enough nitrogen at each site to support 200 bison and the existing elk populations. For reference, an elk needs 13.4 kg of N each year. After considering existing elk populations, is there enough nitrogen at each site to support the reintroduction of 200 bison?